GAGE DEZOORT

GitHub | Homepage | LinkedIn | jgdezoort@gmail.com

SUMMARY

Machine learning researcher with a Princeton Physics Ph.D. specializing in scientific ML, graph transformers, training stability, and efficient model scaling. Available for machine learning research roles in the tri-state area.

RESEARCH EXPERIENCE

Postdoctoral Research Associate — Princeton University, ORFE (2023—Present)

- Developed and validated GNN and graph transformer parameterizations enabling learning rate transfer from ~10M-parameter models to ~0.5B-parameter models, with a principled parameterization for weight decay.
- Built reusable PyTorch Lightning / PyG training pipelines on Princeton GPU clusters, supporting streaming data, on-the-fly graph construction, resumable training, hyperparameter sweeps, and forward/backward pass stability analysis.
- Designed Coupled Particle-Edge Networks with coupled particle-hyperedge attention and derived their hyperparameter transfer parameterizations; engineered a pre-training pipeline for the 100M-jet JetClass dataset and evaluated downstream fine-tuning on TopTagging.
- Mentored student research in GNN-based tracking and graph transformer data augmentation; organized Princeton's Machine Learning in Physics Seminar Series; developed and taught hands-on ML curricula to hundreds of Princeton and UC Irvine students; led a year-long machine learning in particle physics reading group.

Graduate Researcher — Princeton University, Physics (2018—2023)

- Developed GNN-based particle tracking pipelines spanning graph construction, edge classification, and DBSCAN track building; used contrastive representations with learned clustering coordinates for one-shot tracking.
- Derived graph-specific He initialization for deep ReLU GNNs and proved depth-induced oversmoothing for fixed message passing operators; validated residual strategies for mitigating oversmoothing and showed they lead to improved time-to-train and trainability at large depth.
- Built and operated a Python/Awkward Array/Coffea analysis pipeline for a CMS Higgs search using the full LHC Run 2 Ultra Legacy dataset.

EDUCATION

Ph.D. in Experimental Particle Physics — Princeton University (2023)
Certificate, Center for Statistics and Machine Learning.

B.S. in Physics and Engineering Science — University of Virginia (2018)
Highest Distinction. Jefferson Scholar and Rodman Scholar.

SELECTED PUBLICATIONS AND PREPRINTS

Hyperparameter Transfer in Graph Neural Networks | arXiv:2607.05017 (2026)
Principles for Initialization and Architecture Selection in GNNs | SIAM J. Math. Data Sci. (2025)
Graph Neural Networks at the Large Hadron Collider | Nature Reviews Physics (2023)
Charged Particle Tracking via Edge-Classifying Interaction Networks | Comput. Softw. Big Sci. (2021)

SELECTED RESEARCH TALKS

Hyperparameter Transfer: A Recipe for Efficient and Robust Scaling | Plenary talk, HAMLET-PHYSICS (2026)
Hyperparameter Transfer for Graph Transformers | Talk, ACAT (2025)

SELECTED TECHNICAL SKILLS

Machine Learning: deep learning, GNNs, graph transformers, attention, representation learning, model scaling, hyperparameter optimization, benchmarking, ablation studies, numerical stability analysis, contrastive learning
Computing: Python, PyTorch, PyTorch Lightning, PyTorch Geometric (PyG), NumPy, pandas, Git, C/C++, Linux/Unix, HPC/GPU workflows

SELECTED HONORS

School of Physical Sciences Visiting Fellowship — UC Irvine (2025—2026)
Kusaka Memorial Prize in Physics — Princeton University (2022)
Departmental Teaching Award — Princeton University (2020)